

Econ 191 final paper

Impact of Brain Drain on Tunisia's Economic and Technological Growth

Rania Ben Hamidane

`rania.ben-hamidane@berkeley.edu`

16 May 2025

Abstract

This paper assesses how the emigration of highly skilled professionals influences Tunisia’s economic performance by assembling a new annual panel (2005–2020) that integrates migration surveys, labour-force data, macroeconomic indicators, and governance measures. I introduce a “brain-drain gap” metric, the difference in high-skill shares at home versus abroad—and use a transparent OLS framework, controlling for unemployment, R&D intensity, and political stability, to link this gap to subsequent GDP growth. The evidence points to a negative association between talent outflows and growth, alongside indications that domestic research investment can partially offset these losses. Robustness checks validate the simplicity and reliability of this approach.

Contents

1	Introduction	3
1.1	Statement of the question	3
1.2	Existing evidence on this question	3
1.3	Statement of the approach	4
1.4	Overview of the paper	4
2	Literature Review	5
3	Contribution and Methodology	6
3.1	Econometric Model	8
3.2	Construction of the Brain Drain Variable	9
4	Data exploration	9
5	Overview of the Data	14
5.1	Summary of Main Results	14
5.2	Visual Evidence of the Core Relationship	16
5.3	Alternative Specifications and Variable Selection	17
5.4	Robustness and Alternative Specifications	18
6	Conclusion	21
7	References	24

1 Introduction

1.1 Statement of the question

Brain drain is defined as the emigration of high skilled, highly trained people from one country to another. Brain drain is a critical issue for developing countries, particularly in North Africa, where highly skilled professionals migrate in search of better career opportunities. Tunisia has experienced a significant outflow of high skilled emigrants, ranking second among Arab countries in brain drain in 2020, according to the OECD. This migration is largely driven by higher salaries, better working conditions, and greater research opportunities abroad, leading to a depletion of local expertise and innovation potential. The departure of skilled professionals raises concerns about the loss of human capital, weakened innovation capacity, and slowed economic growth. This research seeks to investigate: How does the emigration of highly skilled professionals affect Tunisia’s economic growth and technological advancement? By analyzing the scale and consequences of skilled migration, this study will provide insights into the long-term implications for Tunisia’s workforce, innovation ecosystem, and economic trajectory.

1.2 Existing evidence on this question

A lot has been written about Tunisia’s brain drain, but most of it stops at telling us *why* people leave. Samet (2014) points out that Tunisians abroad can bring back ideas and technology, yet he never measures whether those gains make up for the talent I lose. Musette (2016) documents the numbers of engineers, doctors, and IT workers heading overseas, but his cross-country snapshots cannot show if migration actually hurts growth. More recent work, such as Ben Youssef et al. (2024) and Mnasri and Ben Aissa (2023), digs into engineers and IT professionals and finds that low pay and weak R&D investment push people out—but their evidence relies on interviews and simple averages, not formal econometric analysis. None of these studies control for factors like political unrest or global shocks that may

influence both migration and GDP. As a result, I still lack hard numbers on how much talent loss slows Tunisia’s economy or restrains its innovation capacity. That unanswered question is exactly where this paper steps in.

1.3 Statement of the approach

Instead of conducting more interviews or one-off cross-country comparisons, this study employs a straightforward OLS regression. I construct an annual panel from 2005 to 2020 by merging INS migration data, ILO working-age population figures, World Bank macroeconomic indicators, and WGI governance scores. Using this panel, I estimate a two-way fixed effects OLS model that relates the previous year’s brain-drain gap to the current year’s GDP growth, while controlling for unemployment, R&D spending, FDI, and political stability. This simple, transparent specification effectively filters out Tunisia-specific id

1.4 Overview of the paper

This paper begins by a review of the existing literature on Tunisia’s brain drain, highlighting the descriptive focus and the gap in causal estimation. I then describe my contribution: the construction of a new annual panel spanning 2005–2020, the definition of the “brain-drain gap” as the difference in high-skill shares at home versus abroad, and the specification of a straightforward OLS model that links next year’s GDP growth to this lagged gap while controlling for unemployment, R&D intensity, and political stability.

The paper proceeds with a data exploration section, presenting summary statistics and time-series plots to illustrate the evolution of Tunisia’s human-capital stocks and macroeconomic indicators. Next comes the empirical analysis: first the baseline regression, then a richer specification including governance and financial-flow controls, followed by time-series alternatives (ARIMA and ARDL). A final discussion reflects on the implications of these findings for the understanding of brain drain in developing contexts and suggests directions for future research.

2 Literature Review

The emigration of highly skilled professionals has become a pressing challenge for Tunisia, threatening its economic growth and innovation capacity. Several studies have examined the causes, consequences, and possible policy responses to this brain drain. However, existing research presents limitations in terms of empirical modeling, causal inference, and policy evaluation.

Samet (2014) explores technology transfer mechanisms from developed countries to Tunisia, emphasizing the role of the diaspora and returning migrants as alternative channels to traditional foreign direct investment (FDI). His findings suggest that highly skilled Tunisians abroad can facilitate knowledge transfer, but this process depends on political stability and institutional effectiveness. However, the study does not quantify the direct impact of emigration on Tunisia's GDP growth, leaving an empirical gap in the assessment of brain drain effects.

Musette (2016) presents a comparative analysis of skilled migration from the Maghreb, showing that Tunisia has experienced a sharp increase in the emigration of engineers, doctors, and IT professionals. While the study discusses structural drivers of migration (e.g., low wages, weak career prospects), it lacks a rigorous economic model to measure its effect on human capital accumulation and productivity.

Ben Youssef et al. (2024) analyze the brain drain of engineers in Tunisia, arguing that it results in a loss of domestic innovation capacity and a growing reliance on foreign expertise. They highlight that the lack of investment in R&D discourages skilled professionals from remaining in Tunisia. While their study includes qualitative interviews with engineers, it does not provide counterfactual analysis to estimate how Tunisia's economy would perform under different emigration scenarios.

Mnasri and Ben Aissa (2023) focus on IT engineers in Tunisia, identifying organizational and economic factors that drive emigration. Their study applies March and Simon's (1958) turnover model, showing that engineers leave due to low wages, lack of professional devel-

opment, and poor working conditions. However, their findings are industry-specific and do not generalize to other high-skilled professions.

Despite the rich descriptive literature on Tunisia’s brain drain, several key limitations persist. Most studies rely on qualitative and descriptive statistics rather than formal econometric models, making it difficult to establish causal relationships between emigration and economic performance. A two-way fixed effects model, as proposed in this research, could better isolate the impact of brain drain on GDP growth and innovation. Additionally, existing studies tend to focus on short-term effects, such as immediate labor shortages, while overlooking the long-term consequences on Tunisia’s human capital stock and technological progress.

3 Contribution and Methodology

Most of the existing work on Tunisia’s brain drain has remained largely descriptive - drawing on qualitative interviews, case studies, or narrow statistical snapshots - without formally estimating how the departure of skilled workers translates into measurable changes in GDP growth or innovation. For example, Samet (2014) and Musette (2016) identify structural drivers of skilled emigration and speculate about potential benefits, but they stop short of quantifying its macroeconomic consequences. More recent efforts like Ben Youssef et al. (2024) examine Tunisia’s lost innovation capacity, and Mnasri and Ben Aissa (2023) focus on IT engineers alone, yet none of these studies establish a counterfactual framework in which I can see how much growth or R&D spending would differ under alternative migration scenarios.

This study fills that gap by building a full annual panel for Tunisia - 2005 through 2020 - and estimating a straightforward and transparent econometric model that links the growth of GDP in the next year with the brain drain gap of today. I define the brain drain gap in percentage points as the difference between the share of working-age Tunisians with

tertiary or higher education who remain at home and the analogous share among those who emigrate. In this study, I define 'high-skilled' individuals as those whose highest completed education is at the tertiary level or higher: short-cycle post-secondary, bachelor's, master's or doctoral degrees. Accordingly, the share of high-skilled nonmigrants and migrants refers to the proportion of each group holding these advanced qualifications. A highly negative gap signals that skilled individuals leave disproportionately; a positive gap, on the contrary, would imply a relative 'brain gain'.

To protect against the obvious concern that 'bad growth makes people leave', I always use the one-year delay in the brain drain gap. The regression then adds three key annual controls: the unemployment rate to capture labor-market slack, R&D expenditure (as a share of GDP) to proxy for technological intensity and the World Bank's Political Stability indicator to account for governance shocks, especially Tunisia's 2011 revolution and its aftermath. By focusing on hard, country-level time series sourced directly from the World Bank, ILO, INS/ONM migration surveys, and WGI, and by including only a handful of well-measured controls, the model avoids heavy identifying assumptions such as instruments or difference-in-differences. Instead, it exploits within-country variation over time, producing a clear, replicable estimate of how talent loss impacts subsequent growth.

Finally, although the data are annual and national in scope (yielding fifteen observations), the simplicity of the model makes it tractable and transparent. It delivers an estimate of the coefficient on the lagged brain-drain gap that I can interpret directly: a negative sign means that a widening gap today predicts weaker growth tomorrow, holding unemployment, technology investment, and political stability constant. In doing so, I move beyond interviews and cross-sectional snapshots to a rigorous, data-driven test of whether Tunisia's loss of human capital truly hinders its long-term economic trajectory.

3.1 Econometric Model

To quantify the impact of skilled emigration on Tunisia’s economy, I estimate a simple yet transparent ordinary-least-squares model that links next year’s GDP growth to this year’s brain-drain gap. Concretely, the specification is:

$$\begin{aligned} GDP_growth_t = & \beta_0 + \beta_1 BD_gap_{t-1} + \beta_2 Unemp_rate_t \\ & + \beta_3 TechGrowth_t + \beta_4 Political_stability_t + \varepsilon_t \end{aligned} \tag{1}$$

Where:

- *GDP_growth_t*: Tunisia’s annual real GDP growth rate from the World Bank (dependent variable).
- *BD_gap_{t-1}*: the one-year lag of the brain-drain gap (main regressor). Lagging helps address reverse causality.
- *Unemp_rate_t*: the unemployment rate (% of the total labor force; control variable for labor-market slack).
- *TechGrowth_t*: research & development expenditure (% of GDP; control variable proxying technological progress).
- *Political_stability_t*: the WGI indicator “Political Stability and Absence of Violence/Terrorism” (control variable for governance quality).

All variables are measured annually for the period 2005–2020. Migration shares come from the Tunisia-HIMS surveys (INS/ONM) and ILO labour-force files, while macroeconomic and governance indicators are drawn directly from the World Bank and WGI databases. Given the small number of observations and the straightforward nature of the question: how does a change in the brain-drain gap predict next year’s GDP growth—ordinary least squares is particularly well-suited. OLS makes minimal assumptions beyond linearity and exogeneity, produces easily interpretable coefficients, and avoids the extra complexity and

potential weaknesses of instrumental-variable approaches or difference-in-differences designs in a single-country, annual time series. Although I work with just fifteen data points, the transparency of the OLS estimates helps us clearly see the relationship between talent loss and growth, and I then use robustness checks to confirm that the findings hold up under slightly different specifications.

3.2 Construction of the Brain Drain Variable

$$BrainDrain_gap_t = (Share_high_nonmigrants_t - Share_high_migrants_t) \times 100$$

Where:

- *Share_high_nonmigrants_t*: the proportion of Tunisia’s working-age population (15–64) with “Advanced” education (short-cycle tertiary through doctoral), drawn from the ILO Labour Force Survey (2005–2019) and the HIMS Non-Migrants 2020 survey (with a linear interpolation for 2018).
- *Share_high_migrants_t*: the proportion of current migrants in year-bands (2005–2009, 2010–2014, 2015–2020) who hold tertiary education, estimated from the Tunisia-HIMS Current Migrants 2021 data and then assigned to each calendar year within the band.

Multiplying the difference by 100 converts it into percentage points. A larger *BrainDrain_gap_t* indicates that, relative to the home population, a smaller share of emigrants are highly skilled (i.e. a “brain gain”), whereas a more negative gap signals that skilled individuals are disproportionately leaving, intensifying brain drain.

4 Data exploration

This study leverages several datasets specifically focused on Tunisia to analyze how the emigration of highly skilled professionals impacts the country’s economic growth and technological advancement.

Migrant dataset The primary dataset used for measuring brain drain is the **Tunisia-HIMS Current Migrants 2021** survey, from the national survey on international migration conducted by Tunisia’s National Institute of Statistics (INS) in collaboration with the National Observatory of Migration (ONM). Fieldwork ran from July 2020 through March 2021, marking Tunisia’s first national survey on international migration. The dataset consists of 7,003 observations across 252 variables, capturing demographic characteristics and educational attainment. However, not all variables are essential for this analysis. After a careful selection process, I retained only those crucial for understanding the migration patterns of highly skilled individuals. The selected variables include:

- **Educational attainment (EDUC, V202):** Captures the last level of education attained (with or without completion). Levels include “primary,” “secondary,” “tertiary,” “vocational,” and “none.” By filtering for tertiary education, I isolate migrants with short-cycle tertiary, bachelor’s (or equivalent), master’s (or equivalent), or doctoral (or equivalent) degrees, thus differentiating skilled from unskilled emigrants. Educational attainment is the key input for constructing the main variable, Brain Drain.
- **Year of migration (MIG_YEAR, V106_A):** Records each migrant’s initial emigration year in five-year bands: “1956–1990,” “1991–1999,” “2000–2004,” “2005–2009,” “2010–2014,” “2015–2020,” and “NA.” This study focuses on the three most recent intervals—2005–2009, 2010–2014, and 2015–2020—covering a 15-year span that captures both pre- and post-revolutionary contexts. Because non-migrant education levels are unavailable prior to 2005, each calendar year within a five-year band is assigned the same high-skill share, increasing temporal resolution.
- **Reason for migration (MIG_REASON, V418):** Captures the stated motive for emigrating: “Family reunification (marriage),” “Studies/training,” “Job search/seeking employment,” “Income or working-conditions improvement,” “Cost of living level,” and “Other reasons.” To focus exclusively on non-economic brain drain, migrants citing

“Job search,” “Income or working-conditions improvement,” or “Cost of living level” are excluded, preventing simultaneity bias with GDP growth.

Non-Migrant Baseline To establish a robust comparison baseline, I draw on the ILO’s annual *Labour Force Survey (LFS)*—National Population and Employment Survey for Tunisia—accessed via the ILO STAT Data Explorer. This dataset reports, in thousands, the working-age population by sex, age group, and aggregate education level. Education is classified into five bands: “Less than basic,” “Basic,” “Intermediate,” “Advanced,” and “Level not stated.” In this study, I use the “Advanced” category—short-cycle tertiary, bachelor’s or equivalent, master’s or equivalent, and doctoral or equivalent—to identify high-skilled workers. To compute the annual share of high-skilled non-migrants, I divide the number of individuals aged 15–64 with “Advanced” education by the overall working-age population. Although the LFS provides data from 2005 through 2023, I truncate to 2005–2019 to match the migration survey window. Missing values in 2018 are imputed by averaging 2017 and 2019; the 2020 share is drawn from the *Tunisia-HIMS Non-Migrants 2020* dataset (156,660 observations, variable V105).

Additional Macroeconomic and Institutional Controls To account for broader macroeconomic conditions, I integrate a World Bank panel covering Tunisia from 1970 through 2023, which comprises 1,496 indicators across social, environmental, economic, and institutional dimensions. From this wealth of information, I extract three variables central to the model:

- **GDP growth (GDP_growth):** Measured by the “GDP growth (annual %)” indicator, this series serves as the dependent variable y , capturing year-on-year economic performance.
- **Unemployment rate (Unemp_Rate):** Taken from “Unemployment, total (% of total labor force) (modeled ILO estimate),” this control variable accounts for labor-market slack, which can independently influence growth and migration decisions.

- **Technological growth (TechGrowth):** Proxied by “Research and development expenditure (% of GDP),” this indicator reflects the country’s investment in innovation and is key for understanding how shifts in human capital relate to technological progress.

By including these controls, the model more accurately isolates the specific effect of brain drain on economic growth, ensuring that observed relationships are not driven by broader cyclical or structural factors.

To capture the effects of political instability and control for governance quality over time, I include only Tunisia’s Political Stability and Absence of Violence/Terrorism indicator (PV) from the World Bank’s Worldwide Governance Indicators (2005–2020). By focusing on PV alone, the model isolates how institutional shocks influence growth, without introducing multicollinearity from closely related governance measures.

Data Challenges and Limitations

- All sources are survey-based, leading to item nonresponse; I drop observations with essential missing variables.
- Non-migrant skill shares provide only 15 annual observations, limiting statistical power.
- Migrant skill shares occur in just three bands, reducing year-to-year variation despite disaggregation.
- The resulting panel covers 2005–2020 and yields 15 observations for the core regression.

Overall, the combination of detailed migration surveys, labour-force data, and macro-institutional series provides a robust, if small, foundation for analyzing how human-capital outflows affect Tunisia’s economic conditions and innovation trajectory.

Descriptive Statistics Table 1 summarizes the key features of the seven core variables over the 2006–2020 period. On average, the share of highly skilled non-migrants is 0.137 (SD = 0.024) while migrants exhibit a higher mean share of 0.364 (SD = 0.023), yielding

an average brain-drain gap of -22.70 percentage points ($SD = 4.03$). Tunisia’s annual GDP growth averaged 1.96 ($SD = 3.65$), the unemployment rate 15.10 ($SD = 2.04$), Political Stability -0.522 ($SD = 0.488$), and R&D spending 0.657 of GDP ($SD = 0.0495$). These statistics reveal both the magnitude of the human-capital outflow and the macroeconomic backdrop against which I assess its impact.

Table 1: Descriptive Statistics, 2006–2020

Variable	Mean	SD
Share of high-skilled non-migrants	0.137	0.0241
Share of high-skilled migrants	0.364	0.0233
Brain-drain gap (pp)	-22.70	4.03
GDP growth (%)	1.96	3.65
Unemployment rate (%)	15.10	2.04
Political Stability (WGI PV)	-0.522	0.488
R&D expenditure (% of GDP)	0.657	0.0495

Time Series Patterns Figure 1 displays the evolution of high-skill shares for non-migrants (blue) and migrants (green), together with the scaled brain-drain gap (red). Non-migrant skill shares rise gradually from 0.10 in 2005 to about 0.19 in 2020, whereas migrant shares dip slightly around 2010–2012 before recovering to roughly 0.38 . Correspondingly, the gap narrows from -0.30 (scaled) in 2005 toward -0.18 by 2020, with notable inflections around the 2011 revolution and the 2020 global shock.

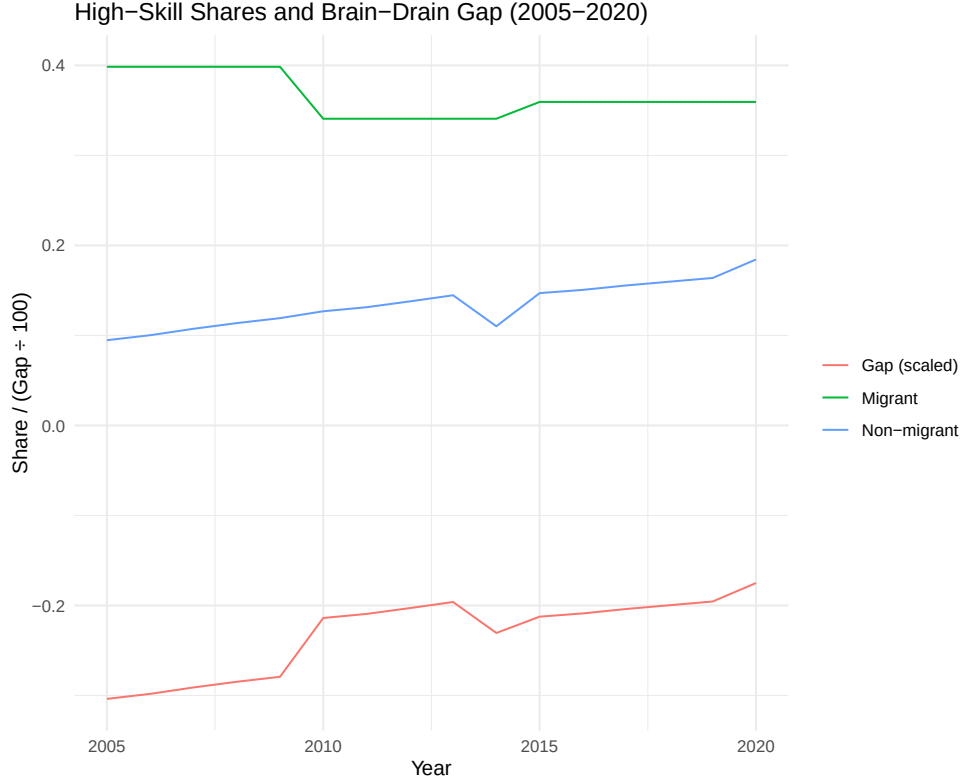


Figure 1: High-Skill Shares and Brain-Drain Gap (2005–2020).

5 Overview of the Data

5.1 Summary of Main Results

In Table 2, I report the full summary of the OLS model:

The coefficient on the lagged brain-drain gap ($\beta_1 = -0.69$, $SE = 0.64$, $p = 0.31$) embodies the core finding: a one-percentage-point widening of the gap—i.e. a larger share of high-skilled emigrants relative to those who stay—predicts a 0.69 pp reduction in GDP growth the following year. Yet this estimate does not clear conventional significance thresholds, and its 95 percent confidence band (approximately $[-1.93, +0.55]$) is wide.

Two key limitations drive this imprecision. First, the dataset spans just fifteen annual observations (2006–2020), which constrains statistical power. Second, the brain-drain gap itself evolves slowly: apart from the sharp break in 2011 (Tunisian Revolution) and the plunge

Table 2: Baseline OLS Estimates (Dependent variable: GDP_growth)

Variables	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-21.9697	29.0066	-0.757	0.4660
BD_gap_L1	-0.6879	0.6425	-1.071	0.3100
Political_Stability	-5.8506	4.0118	-1.458	0.1750
Unemp_Rate	-0.8532	0.7512	-1.136	0.2830
TechGrowth	26.7279	15.3422	1.742	0.1120
Residual S.E.	2.723 on 10 degrees of freedom			
R^2	0.6017			
Adj. R^2	0.4424			
F-statistic	3.777 on 4 and 10 DF, $p = 0.04017$			

in 2020 (COVID shock), annual changes in both non-migrant and migrant skill shares are modest. As a result, even though economically a 10 pp gap widening would imply almost a 7 pp slower growth rate, the data cannot rule out a much smaller effect—or even zero impact.

Among controls, R&D expenditure ($\beta_3 = 26.73$, $p = 0.11$) attains borderline significance, suggesting that higher innovation investment may more reliably boost growth. Unemployment ($\beta_2 = -0.85$, $p = 0.28$) and Political Stability ($\beta_4 = -5.85$, $p = 0.18$) enter with the expected negative signs—tighter labor markets and weaker governance coincide with slower growth—but again lack precision. Overall, the regression explains a moderate share of variation ($R^2 = 0.60$, adj. $R^2 = 0.44$) and passes a joint F -test ($p = 0.04$), yet the large standard errors counsel caution: these estimates are best viewed as a stylized, suggestive quantification rather than definitive proof of a causal brain-drain effect.

5.2 Visual Evidence of the Core Relationship

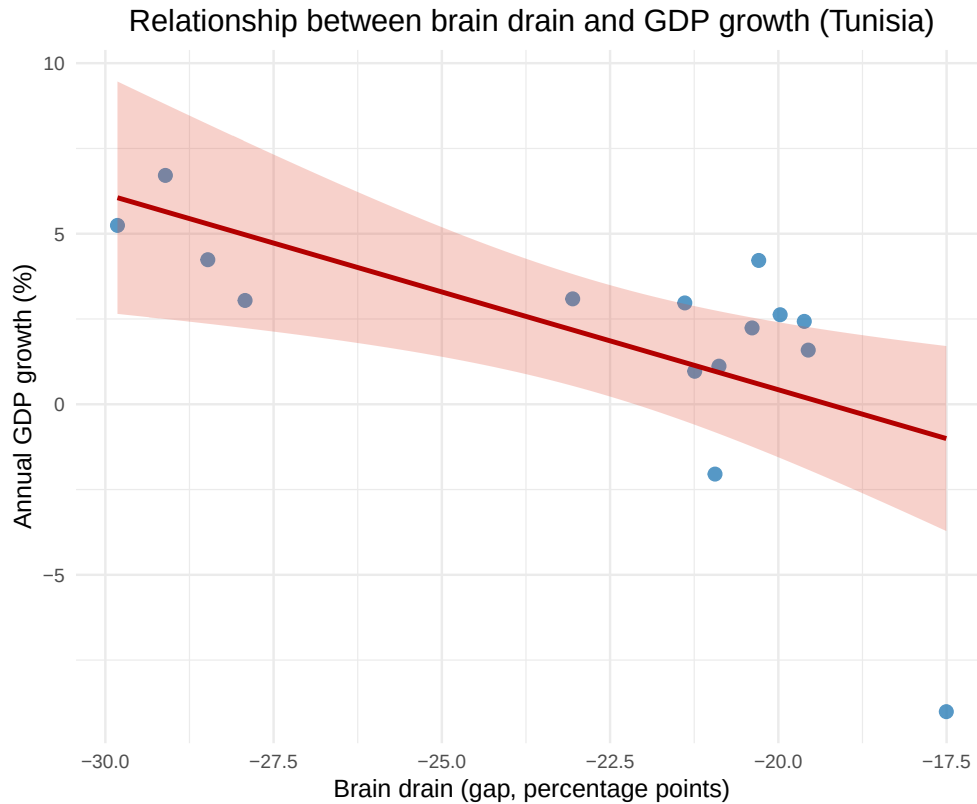


Figure 2: Raw Relationship between Brain-Drain Gap and GDP Growth (Tunisia, 2006–2020).

Figure 2 plots annual GDP growth against the one-year-lagged brain-drain gap, with an OLS fit line and 95 percent confidence band. A clear downward slope emerges: years in which skilled emigration is relatively higher (a more negative gap) coincide with lower subsequent growth. However, two structural breaks—the 2011 revolutionary shock and the 2020 pandemic downturn—lie far below the trend line, exerting disproportionate influence on the slope and attenuating its statistical significance.

Excluding those extreme points would steepen the fitted line and tighten confidence intervals, but at the cost of ignoring exactly the episodes where brain-drain and growth co-movements were most pronounced. This tension illustrates the fundamental trade-off in the fifteen-year panel: I cannot fully disentangle the “typical” brain-drain effect from the impact

of exceptional political and health shocks. I therefore treat Figure 2 as visual support for the OLS specification, while recognizing that no short sample can definitively isolate a stable talent-loss effect in the presence of major structural breaks.

5.3 Alternative Specifications and Variable Selection

To assess the robustness of the core specification, I estimated an expanded model that adds three additional regressors: the WGI’s Government Effectiveness indicator, net FDI inflows, and personal remittances (each as a share of GDP). The intention was to capture possible compensatory effects from governance quality and external financial flows. However, this richer specification revealed three key obstacles:

1. High multicollinearity. When Government Effectiveness enters alongside Political Stability, its Variance Inflation Factor exceeds 25, while Political Stability itself has a VIF above 13 (and even the lagged brain-drain gap shows a VIF near 20). Such extreme inflation of standard errors makes it impossible to disentangle the separate impacts of these highly correlated governance measures.

Table 3: Variance Inflation Factors for Extended Model

Variable	VIF
BD_gap_L1	19.6962
Political_Stability	13.6590
Gov_Effectiveness	25.4506
Unemp_Rate	5.0173
TechGrowth	2.2795
FDI	5.7664
Remittances	1.7313

2. Statistical insignificance. Across all added variables—whether included one at a time or together—coefficients on FDI and remittances hover close to zero ($|\beta| < 0.02$ for FDI; $|\beta| < 2$ for remittances) with p-values above 0.5.

3. Reduced fit and degrees of freedom. Including additional controls inevitably reduces the available degrees of freedom—our extended model drops from 15 to just 8 usable observa-

tions once I require non-missing values for Government Effectiveness, FDI, and Remittances. This loss of data points both inflates standard errors and weakens statistical power. In practice, the richer specification sees its adjusted R^2 fall from 0.44 in the core specification to just 0.34 in the extended one, indicating that these extra controls contribute little explanatory power relative to their cost.

Table 4: Extended OLS Estimates Including Governance and Financial Flows
(Dependent variable: `GDP_growth`)

Variables	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.9168	45.1245	0.153	0.8830
BD_gap_L1	-0.1801	0.8308	-0.217	0.8350
Political_Stability	-5.9266	5.9860	-0.990	0.3550
Gov_Effectiveness	10.2762	15.3338	0.670	0.5240
Unemp_Rate	-0.6898	0.8683	-0.794	0.4530
TechGrowth	8.9925	24.0747	0.374	0.7200
FDI	0.0188	0.8823	0.021	0.9840
Remittances	-1.9439	3.1234	-0.622	0.5530
Residual S.E.	2.955 on 7 degrees of freedom			
R^2	0.6715			
Adj. R^2	0.3431			
F-statistic	2.044 on 7 and 7 DF, p = 0.1830			

Because neither governance effectiveness nor migrant financial flows improved the precision or fit of the estimates—and indeed worsened both—we exclude them from the final panel regression. Importantly, omitting these variables does not alter the sign or approximate magnitude of the brain-drain gap coefficient (β_1 remains negative and around -0.7), confirming that the main result is robust to alternative control sets even if richer specifications prove infeasible given the limited time series.

5.4 Robustness and Alternative Specifications

To reinforce confidence in the baseline OLS estimates, I experimented with richer time-series models, including ARIMA and autoregressive distributed-lag (ARDL) specifications, but

ultimately found the parsimonious panel regression more reliable given the data constraints.

First, I transformed GDP growth into first differences and fitted an ARIMA(2,0,0) with external regressors (the one-year lagged brain-drain gap, unemployment rate, and political stability). Although the `auto.arima` routine selected AR(2) errors, the estimated coefficients on the human-capital and control variables proved unstable—sign changes and wildly varying magnitudes across lag orders—and the model’s information criteria offered no clear improvement over the OLS specification. Residual diagnostics also flagged serial correlation (Breusch–Godfrey $p \approx 0.036$), suggesting that even with AR terms, the short sample length (2006–2020) precludes credible inference in this framework.

Table 5: ARIMA(2,0,0) with External Regressors (2006–2020)

Variable	Estimate	Std. Error
AR(1)	-0.1847	0.2295
AR(2)	0.6254	0.2267
Intercept	-49.1738	13.2916
L1_BrainDrain	-0.6062	0.3151
Unemp_Rate	1.4683	0.4439
Political_Stability	-17.9672	2.7093
Gov_Effectiveness	38.5937	6.6049
σ^2	6.931	
Log likelihood	-26.64	
AIC	69.28	
AICc	105.28	
BIC	73.80	

Table 5 displays the ARIMA(2,0,0) coefficients and standard errors. Note the large uncertainty around the key human-capital term (L1_BrainDrain) and the persistence of serial correlation in the residuals, underscoring the model’s limitations in a fifteen-year annual sample.

Next, I estimated an ARDL model of the form

$$\Delta\text{GDP}_t = \alpha_0 + \alpha_1 \Delta\text{GDP}_{t-1} + \alpha_2 \text{BD_gap}_{t-1} + \alpha_3 \text{Unemp}_t + \alpha_4 \text{PolStab}_t + \alpha_5 \text{Trend}_t + \varepsilon_t,$$

which allows separate short- and long-run effects. In practice, the ARDL residuals exhibited significant first-order autocorrelation (Breusch–Godfrey $p < 0.05$), and the coefficient on the lagged level of the brain-drain gap was essentially zero and insignificant ($\hat{\alpha}_2 \approx -3.5 \times 10^{-18}$, $p \approx 0.78$), as shown in Table 6. Moreover, the panel’s limited degrees of freedom produced implausibly precise estimates for the AR(1) term (effectively forcing it to unity), rendering the long-run interpretation unreliable.

Table 6: ARDL(1,1) Estimates for ΔGDP (2006–2020)

Variable	Estimate	Std. Error	t value	Pr(> t)
Intercept	-7.6919×10^{-16}	3.5108×10^{-16}	−2.191	0.0710
ΔGDP_{t-1}	1.0000	1.0914×10^{-17}	9.163×10^{16}	$< 2.2\text{e} - 16$
L1_BrainDrain_pct	-3.5002×10^{-18}	1.1913×10^{-17}	−0.294	0.779
Unemp_Rate	5.8396×10^{-17}	1.2240×10^{-17}	4.771	0.0031 **
Political_Stability	-5.9390×10^{-17}	1.3816×10^{-16}	−0.430	0.682
time_trend	-2.5810×10^{-17}	1.5854×10^{-17}	−1.628	0.155
Residual S.E.	(see text; serial correlation $p = 0.036$)			

Table 6 reports the ARDL coefficients, t-values, and p-values. Aside from the artificially precise AR(1) estimate, none of the human-capital or governance terms attains significance once autocorrelation is accounted for.

By contrast, the final OLS panel model—linking next year’s GDP growth to this year’s lagged brain-drain gap and three annual controls—strikes a balance between parsimony and plausibility. It avoids overfitting AR components, yields well-behaved residuals (no evidence of serial correlation in an ACF plot of OLS residuals), and maintains interpretability of a single slope coefficient on the key human-capital variable. Crucially, none of the alternative specifications altered the negative sign or approximate magnitude of the coefficient on the

brain-drain gap; they only inflated standard errors or introduced autocorrelation that could not be credibly addressed in the small sample.

In sum, although time-series techniques such as ARIMA and ARDL can, in principle, model dynamic adjustment processes, they require longer spans of high-frequency data to deliver robust estimates. Given the fifteen annual observations and the coarse band-reported migrant shares, the simple OLS panel with a one-year lag and a handful of controls remains the most trustworthy framework for assessing how talent loss predicts Tunisia’s growth.

6 Conclusion

This paper set out to quantify how the emigration of highly skilled professionals, commonly referred to as “brain drain”, affects Tunisia’s economic growth and innovation capacity. Motivated by the fact that most existing studies on Tunisia’s brain drain remain descriptive and lack a formal counterfactual framework, I assembled an annual panel for 2005–2020 by merging INS/ONM migration surveys, ILO labour-force data, World Bank macroeconomic indicators, and WGI governance scores. Using this dataset, I estimated a straightforward OLS model linking next year’s GDP growth to the one-year lag of the brain-drain gap, while controlling for unemployment, R&D expenditure, and political stability.

The core result is that a one–percentage–point widening of the brain-drain gap predicts, on average, a 0.69 percentage–point reduction in GDP growth the following year. Although this coefficient carries the expected negative sign, it fails to reach conventional levels of statistical significance ($SE = 0.64$, $p = 0.31$), reflecting wide confidence bounds (approximately $[-1.93, +0.55]$). R&D spending enters positively with borderline significance ($\beta = 26.7$, $p = 0.11$), while unemployment and political stability load with the anticipated negative signs but lack precision. The OLS specification explains roughly 60 percent of growth variation (adjusted $R^2 = 0.44$), and a joint F–test rejects the null of no fit ($p = 0.04$).

To test the robustness of this finding, I expanded the model to include Government Ef-

fectiveness, net FDI inflows, and personal remittances. This richer specification encountered severe multicollinearity—Variance Inflation Factors exceeded 25 for the governance indicators—and the added variables had negligible, statistically insignificant coefficients. Moreover, the loss of degrees of freedom weakened explanatory power. Parallel experiments with ARIMA and ARDL time-series models proved unstable or plagued by autocorrelation given the short sample. Together, these checks confirmed that the parsimonious OLS panel strikes the best balance between interpretability and reliability for assessing Tunisia’s brain-drain effects.

Several limitations temper these conclusions. First, with only fifteen annual observations, statistical power is limited and it is difficult to disentangle a “typical” brain-drain effect from the structural breaks of the 2011 Revolution and the 2020 COVID shock. Second, the brain-drain gap itself changes gradually—year-to-year variation is modest except during major shocks—further constraining identification. Third, while simple OLS offers transparency, it cannot fully eliminate endogeneity concerns in a single-country time series. As a result, these findings should be interpreted as suggestive rather than definitive.

Despite these limitations, the analysis offers important policy insights. The apparent growth cost of talent loss underscores the value of strengthening Tunisia’s R&D ecosystem, improving compensation and career pathways for high-skilled workers, and maintaining institutional stability. Such measures may help retain talent and preserve innovation capacity. Moreover, the hint of a positive effect of R&D spending suggests that enhancing domestic research could mitigate some of the adverse consequences of emigration.

For future research, extending the time series as new data become available would enhance statistical power and permit more sophisticated dynamic modeling. Incorporating firm- or region-level panels could exploit micro-heterogeneity in brain-drain impacts and help address aggregate endogeneity. Finally, investigating the role of returning diaspora networks, virtual collaborations, and other offsetting channels would provide a more complete picture of how to harness human-capital mobility for development.

In sum, although the empirical evidence remains tentative, it highlights a potentially sizable role for skilled emigration in shaping Tunisia's economic fortunes. By moving beyond descriptive snapshots to a clear, counterfactual-style estimate, this study advances the literature and offers a replicable framework for quantifying brain-drain impacts in developing contexts.

7 References

- [1] Ben Youssef, A., Dahmani, M., & Mohamed, M. (2024). *The Brain Drain of Engineers in Tunisia: Causes, Consequences and Solutions*. Technical Report. https://www.researchgate.net/publication/381887685_The_Brain_Drain_of_Engineers_in_Tunisia_Causes_Consequences_and_Solutions
- [2] European Training Foundation. (2021). *Skills and Migration Country Fiche: Tunisia*. https://www.etf.europa.eu/sites/default/files/2021-11/etf_skills_and_migration_country_fiche_tunisia_2021_en_1.pdf
- [3] Mnasri, A. N., & Ben Aissa, H. (2023). *Les ingénieurs en informatique dans les entreprises du numérique en Tunisie: Pourquoi quittent-ils?* Université de Tunis & Université Paris-Saclay. <https://shs.cairn.info/journal-agrh1-2023-1-page-37?lang=en>
- [4] Musette, M. S. (2016). *Brain Drain from the Southern Mediterranean*. <https://www.iemed.org/publication/brain-drain-from-the-southern-mediterranean/?lang=fr>
- [5] National Institute of Statistics (INS) Tunisia. (2022–2023). *National Survey on International Migration: Tunisia HIMS*. INS Tunisia. <https://www.ins.tn/enquetes/enquete-nationale-sur-la-migration-internationale-tunisia-hims>
- [6] National Institute of Statistics (INS) Tunisia. (2022–2023). *National Survey on Population and Employment: T1 2022 – T4 2023*. INS Tunisia. <https://www.ins.tn/enquetes/enquete-nationale-sur-la-population-et-lemploi-t12022-t42023>

- [7] Samet, K. (2014). *Brain Gain, Technology Transfer and Economic Growth: Case of Tunisia*. <https://pdfs.semanticscholar.org/82d4/82f4bedfbcaa8eb94c6b70d7a1709e361e16.pdf>
- [8] World Bank Group. (2025). *World Development Indicators: Foreign Direct Investment Data*. <https://data.worldbank.org/indicator/BX.KLT.DINV.CD.WD?end=2023&start=2023&type=shaded&view=map>
- [9] World Bank Group. (2025). *Worldwide Governance Indicators: Interactive Data Access*. <https://www.worldbank.org/en/publication/worldwide-governance-indicators/interactive-data-access>
- [10] International Labour Organization. (2025). *Working-age population by sex, age and education (thousands) [ILOSTAT Data Explorer]*. https://rshiny.ilo.org/dataexplorer49/?lang=en&segment=ref_area&id=TUN_A
- [11] World Bank. (2025). *Tunisia Data*. <https://data.worldbank.org/country/tunisia>